Cooperative Robots for Smart Factories

An open-source logistics integration blueprint

iisys
Institut für Informationssysteme der Hochschule Hof

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Abstract

TurtleBot 4 and MyCobot 280 cooperate with a Siemens smart factory. Modular software connects navigation, vision and manipulation using onboard sensors, without external tracking or proprietary middleware.

System & methods

TurtleBot 4: ROS 2 Jazzy, Nav2 and AMCL for mapping, localisation and delivery.

MyCobot 280: Jetson Nano, MoveIt 2 and camera-based product recognition.

Factory gateway: OPC UA links the robotics stack to Siemens S7-1500 PLCs. A shared dashboard coordinates operation.



B002 Future Lab: robots, team and factory testbed.

Three logistics missions

Product delivery
Identify the product, collect it and deliver it to the destination for its colour.

Defective-component retrieval
Transport rejected components to an inspection area.

Material restocking
Bring supplies to the factory inventory station.



Research collaboration at Hof University.

Delivery cycle: detect → pick up → hand off → deliver → return to dock

Contribution & next steps

- Replaceable modules connect heterogeneous robotics and industrial systems.
- Coordinated handoffs exchange product class, arrival and release status.
- Next: benchmark complete cycles in clear, static and dynamic obstacle scenarios.

Hochschule Hof
University of Applied Sciences

Institute for Information Systems
Hof University of Applied Sciences
Alfons-Goppel-Platz 1 · 95028 Hof
www.hof-university.de

Supervision: Prof. Dr.-Ing. Christoph Koch
Future Lab: Shantall Marucia Cisneros Saldana

Matheus Hipólito Carvalho
EESC–USP / Hof University of Applied Sciences
matheus.carvalho@hof-university.de